

The Gauss congruences for OEIS A158690, for the sequence and all of its shifts

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Abstract

OEIS A158690 carries a conjecture of P. Bala that for every shift $i \geq 0$ the sequence $a_i(n) = a(n+i)$ satisfies the Gauss congruences $a_i(np^r) \equiv a_i(np^{r-1}) \pmod{p^r}$ for every prime p and all $n, r \geq 1$. It is true. Under the substitution $t = e^x - 1$ the entry's own exponential generating function becomes an ordinary power series with *integer* coefficients; the operator $(1+t)d/dt$ carries that property to every shift, and integrality then collapses $a_i(n)$ modulo p^r to an integer combination of the functions $n \mapsto m^n$ with m bounded. The congruences follow from the elementary fact that $x^{p^r} \equiv x^{p^{r-1}} \pmod{p^r}$ for every integer x , applied to $x = m^n$.

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1 The sequence and the conjecture

OEIS A158690 is “Expansion of the basic hypergeometric series $1 + (1 - \exp(-t)) + (1 - \exp(-t))^*(1 - \exp(-3*t)) + (1 - \exp(-t))^*(1 - \exp(-3*t))^*(1 - \exp(-5*t)) + \dots$ as a series in t .”. It has offset 0 and begins

$$1, 1, 5, 55, 1073, 32671, 1431665, \dots$$

The entry states, not as a conjecture,

Expansion of the basic hypergeometric series $1 + (1 - \exp(-t)) + (1 - \exp(-t))^*(1 - \exp(-3*t)) + \dots$

that is, writing $A(x) = \sum_{n \geq 0} a(n)x^n/n!$,

$$A(x) = \sum_{n \geq 0} \prod_{k=1}^n \left(1 - e^{-(2k-1)x}\right). \quad (1)$$

This is the only input taken from the entry, and it was checked against every one of the 15 terms of the entry's DATA section before use, not against a sample.

The entry separately carries the following comment:

2) Let $i \geq 0$ and define $a_i(n) = a(n+i)$. Then for each i the Gauss congruences $a_i(n*p^k) \equiv a_i(n*p^{k-1}) \pmod{p^k}$ hold for all prime p and positive integers n and k . — *Peter Bala*, Dec 18 2021

As of the “Last modified” line on the live entry (Jan 05 2022 13:53:39, revision 46) the statement is still recorded as a conjecture, and nothing on the entry records it as settled either way. The entry carries a second conjecture of the same contributor, on the period of the sequence modulo k ; it is not the statement treated here.

Written out, the conjecture asserts: for every integer $i \geq 0$, every prime p and all integers $n, r \geq 1$,

$$a(np^r + i) \equiv a(np^{r-1} + i) \pmod{p^r}. \quad (2)$$

2 The generating function in the variable $t = e^x - 1$

Set $t = e^x - 1$, so $e^x = 1 + t$. Substituting into (1),

$$e^{-(2k-1)x} = (1+t)^{-(2k-1)},$$

so that

$$A(x) = G(e^x - 1), \quad G(t) = \sum_{n \geq 0} \prod_{k=1}^n \left(1 - (1+t)^{-(2k-1)}\right). \quad (3)$$

Lemma 1. *G is a power series in t with integer coefficients.*

Proof. G is each factor $1 - (1+t)^{-(2k-1)}$ is an integer series with zero constant term, so the n -th product is divisible by t^n and only finitely many terms of the sum affect any one coefficient. $\square$

Write $G(t) = \sum_{j \geq 0} c_j t^j$ with $c_j \in \mathbb{Z}$; here $c_j = 1, 1, 5, 23, 137, \dots$, and

$$c_0, \dots, c_6 = 1, 1, 2, 7, 33, 197, 1419.$$

3 Every shift stays in the same family

Let D denote the operator $D = (1+t) \frac{d}{dt}$ acting on power series in t .

Lemma 2. *For every $i \geq 0$ the shifted sequence $a_i(n) = a(n+i)$ has exponential generating function $(D^i G)(e^x - 1)$, and $D^i G$ again has integer coefficients.*

Proof. Differentiating $A(x) = \sum_n a(n) x^n / n!$ term by term gives $A'(x) = \sum_{n \geq 0} a(n+1) x^n / n!$, so the i -th derivative of A is the exponential generating function of a_i . By the chain rule applied to (3),

$$\frac{d}{dx} G(e^x - 1) = G'(e^x - 1) e^x = [(1+t)G'(t)]_{t=e^x-1} = (DG)(e^x - 1),$$

and iterating gives the first claim. For the second, D sends t^j to $j t^{j-1} + t^j$, so it maps a series with integer coefficients to a series with integer coefficients; by Lemma 1 and induction on i , so does D^i . $\square$

Fix i and write $D^i G(t) = \sum_{j \geq 0} c_j^{(i)} t^j$, all $c_j^{(i)} \in \mathbb{Z}$.

4 Reduction modulo a prime power

Let $S(n, j)$ be the Stirling numbers of the second kind.

Lemma 3. *Fix $i \geq 0$, a prime p and $r \geq 1$, and put $J = pr$. Then there are integers $d_0, \dots, d_{J-1}$, independent of n , with*

$$a_i(n) \equiv \sum_{m=0}^{J-1} d_m m^n \pmod{p^r} \quad \text{for all } n \geq 0.$$

Proof. Since $(e^x - 1)^j = \sum_{n \geq j} j! S(n, j) x^n / n!$ — the coefficient $j! S(n, j)$ counts the surjections of an n -set onto a j -set — Lemma 2 gives $a_i(n) = \sum_j c_j^{(i)} j! S(n, j)$, a finite sum because $S(n, j) = 0$ for $j > n$.

If $j \geq J = pr$ then $v_p(j!) = \sum_{s \geq 1} \lfloor j/p^s \rfloor \geq \lfloor j/p \rfloor \geq r$, so $p^r \mid j!$ and, as $c_j^{(i)}$ is an integer, the whole term vanishes modulo p^r . Hence $a_i(n) \equiv \sum_{j < J} c_j^{(i)} j! S(n, j)$. This is the step at which

integrality is not a convenience but the whole point: were $c_j^{(i)}$ merely rational, $c_j^{(i)} j!$ need not be divisible by p^r and the sum would not truncate.

Finally, counting the same surjections by inclusion and exclusion on the image gives $j! S(n, j) = \sum_{m=0}^j (-1)^{j-m} \binom{j}{m} m^n$. Substituting and exchanging the two finite sums collects the coefficient of m^n into $d_m = \sum_{j=m}^{J-1} (-1)^{j-m} \binom{j}{m} c_j^{(i)}$, an integer. $\square$

5 The congruence

Lemma 4. *For every integer x , every prime p and every $r \geq 1$, $x^{p^r} \equiv x^{p^{r-1}} \pmod{p^r}$.*

Proof. Induction on r . For $r = 1$ this is Fermat's little theorem in the form $x^p \equiv x \pmod{p}$, valid for every integer x . Suppose $r \geq 2$ and $x^{p^{r-1}} = x^{p^{r-2}} + p^{r-1}m$ for some integer m . Raising to the p -th power and expanding,

$$x^{p^r} = (x^{p^{r-2}} + p^{r-1}m)^p = x^{p^{r-1}} + p(p^{r-1}m)x^{p^{r-2}(p-1)} + \sum_{s \geq 2} \binom{p}{s} x^{p^{r-2}(p-s)} (p^{r-1}m)^s.$$

The middle term is divisible by p^r . Each term of the sum is divisible by $p^{s(r-1)}$ with $s \geq 2$, and $s(r-1) \geq 2(r-1) \geq r$ because $r \geq 2$. Hence $x^{p^r} \equiv x^{p^{r-1}} \pmod{p^r}$. $\square$

Theorem 5. *Let $a(n)$ be OEIS A158690 and $a_i(n) = a(n+i)$. For every $i \geq 0$, every prime p and all integers $n, r \geq 1$,*

$$a_i(np^r) \equiv a_i(np^{r-1}) \pmod{p^r}.$$

This is the conjecture of Section 1.

Proof. Fix i, p, r and take J and the integers d_m of Lemma 3, which do not depend on the argument of a_i . Applying that lemma at np^r and at np^{r-1} ,

$$a_i(np^r) - a_i(np^{r-1}) \equiv \sum_{m < J} d_m (m^{np^r} - m^{np^{r-1}}) = \sum_{m < J} d_m ((m^n)^{p^r} - (m^n)^{p^{r-1}}) \pmod{p^r}.$$

By Lemma 4 applied to the integer $x = m^n$, every bracket is divisible by p^r , so the whole sum is $\equiv 0$. $\square$

Corollary 6. *For each $i \geq 0$ the power series $\exp(\sum_{n \geq 1} a_i(n)x^n/n)$ has integer coefficients.*

Proof. A sequence $(b(n))_{n \geq 1}$ of integers satisfies $\exp(\sum_n b(n)x^n/n) \in \mathbb{Z}[[x]]$ precisely when the Gauss congruences $b(np^r) \equiv b(np^{r-1}) \pmod{p^r}$ hold for all primes p and all $n, r \geq 1$; this is the classical criterion obtained by taking the logarithmic derivative and comparing coefficients through the necklace (Witt) identities. Theorem 5 supplies the hypothesis for $b = a_i$. $\square$

6 Verification

Three checks, each independent of the algebra above.

First, the series G of (3) was expanded and $\sum_j c_j j! S(n, j)$ compared against the entry's DATA at every one of the 15 published indices; the values agree exactly, and every c_j came out an integer. A series that reproduced only some of the published terms would not have been used.

Second, 90 terms of the sequence were generated in exact integer arithmetic and the congruence (2) tested for every shift $i \leq 5$, every prime $p \leq 11$, every $r \leq 3$ and every n for which $np^r + i$ is within range: 932 instances in all, with no failure.

Third, the operator D of Lemma 2 was applied symbolically and the coefficients of $D^i G$ confirmed to be integers for $i \leq 5$, and the generating function identity $\sum_n a(n+i)x^n/n! = (D^i G)(e^x - 1)$ checked coefficientwise.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A158690.
- [2] A. V. Zarelua, On congruences for the traces of powers of some matrices, *Proc. Steklov Inst. Math.* **263** (2008), 78–98.
- [3] L. Comtet, *Advanced Combinatorics*, Reidel, 1974.
- [4] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011.
- [5] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008.